

Service Priority AI

Responsible Azure MLOps for public-service case prioritisation

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30 June 2026 | Virtual interview

Context And My Role

Why the system exists, who uses it, and what I personally built

Business problem

Urgent cases can be hidden in mixed service queues.

Officers need consistent triage support, not an automated decision maker.

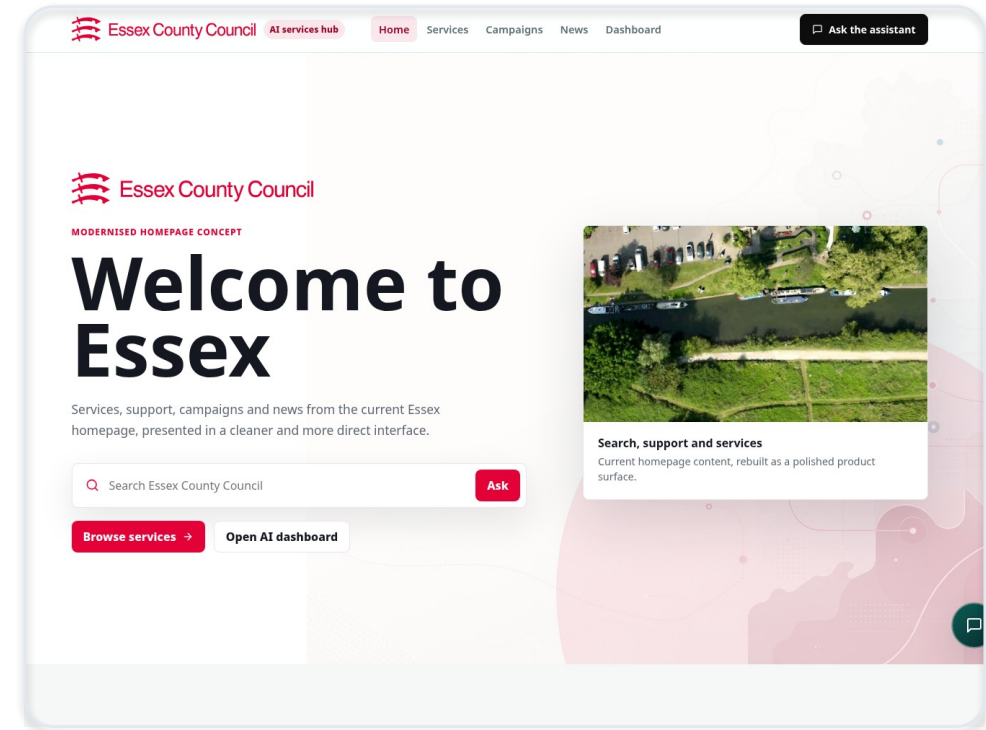
Managers need visibility of risk, demand, overrides, and model behaviour.

Users

Case officers reviewing service requests.

Service managers monitoring queues and exceptions.

Information Governance and responsible AI reviewers.



My contribution

Designed and built the ML lifecycle, API boundary, Azure deployment assets, monitoring outputs, and governance documentation.

Data And Real-World Limitations

The data was useful for proving the pattern, but the risks are the point

TRAINING ROWS

20k

synthetic service cases

VALIDATION ROWS

5k

hold-out evaluation

DECISION ROLE

Advisory

human review required

Inputs

Service type, SLA, district, channel, days open, previous contacts.

Vulnerability flag, deprivation band, accessibility need.

Urgency notes transformed with explainable text features.

Limitations

Synthetic labels do not prove live resident-service performance.

Real notes would be sensitive, noisy, incomplete, and unevenly written.

Vulnerability and deprivation can create proxy-risk concerns.

Controls

DPIA-lite, model card, responsible AI assessment.

No raw real resident data in the demo.

Fairness slices, drift review, override monitoring, and human accountability.

Model And Evaluation

Transparent baseline first, because missed urgent cases are the service risk

Baseline model

scikit-learn LogisticRegression pipeline.

ColumnTransformer over categorical, numeric, boolean, and TF-IDF text features.

Chosen for repeatability, explainability, and public-sector accountability.

ACCURACY

90.2%

validation

MACRO F1

90.35%

balanced view

HIGH RECALL

94.08%

harm-oriented

Validation checks

Hold-out metrics, confusion matrix, class recall, confidence review, fairness slices.

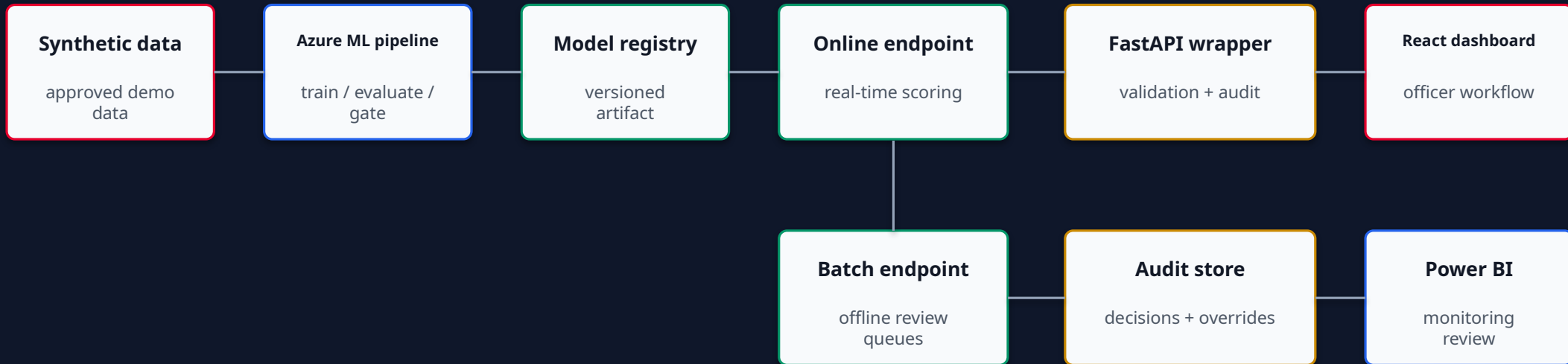
High-priority recall gate passed; fairness remains review_required.

The screenshot shows a web interface for 'Essex County Council' under the 'EMPLOYEE WORKSPACE' and 'Case details' sections. The case is titled 'Housing repair' and is marked as 'high priority'. The description states 'Fire risk and also children in house' with a 'Within 2 h' time indicator. The 'Case summary' section provides a brief overview and access instructions. The 'Assigned officer' is Alice Morgan. A 'Variables' table lists case details, and an 'Evidence' section includes a 'Repair photo'.

Case reference	ECC-365-1042	Service	Housing repair
Team	Contact centre	Status	In review
Due	Within 2 hours	Source	Outlook shared mailbox
District	Chelmsford	Channel	phone
Days open	5	Prior contacts	1
Accessibility need	Yes	Vulnerability context	Yes
Duplicate signal	No	Area context	high

Azure Architecture And Service Choices

Governed serving path with online and batch scoring



Why Azure ML

Pipelines, pinned environments, registry, managed endpoints, traffic control, and reproducibility.

Why API wrapper

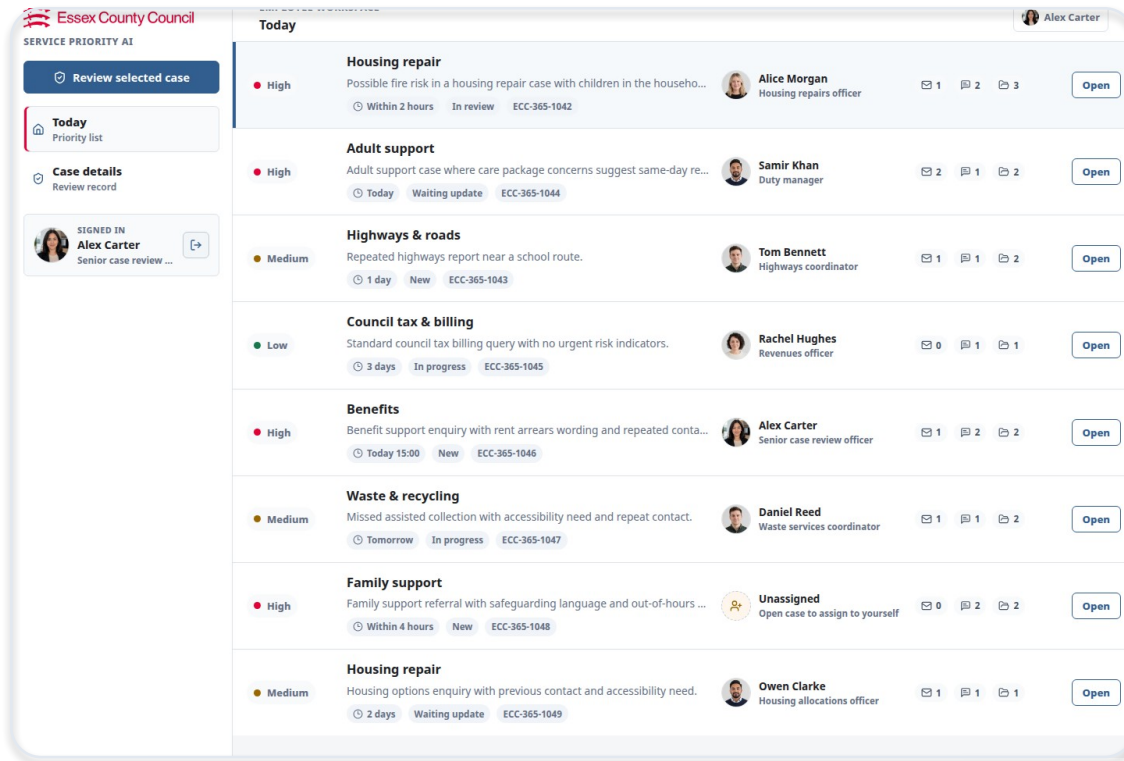
Browser never handles Azure ML endpoint secrets; the API centralises validation, logging, and audit.

Why batch scoring

Supports scheduled review queues, monitoring backfills, and retrospective evaluation.

Production Safety Controls

The system is designed to be supportable before it is impressive



Engineering

Typed FastAPI and Pydantic contracts.
Backend tests, frontend build check, local validation script.
Versioned artifacts, registry tags, and model metadata.

Governance

Model card, DPIA-lite, responsible AI assessment.
Human review, override reasons, and audit trail.
Fairness status remains review_required before real deployment.

Operational controls

Azure Table Storage audit pattern, Application Insights/Azure Monitor health signals, endpoint rollback by traffic/deployment, small compute, scale-to-zero, and monthly budget alerts.

Monitoring, Incident And Learning

How I would know the service was degrading and what changed after a rollout issue

Monitoring

API health, request volume, error rate, p95 latency, model-loaded state.

Class mix, confidence, low-confidence rate, explanation coverage.

Drift, fairness slices, high-priority recall, audit completeness, and overrides.

Incident

Linux App Service plan creation in UK South was blocked by subscription VM quota.

Root cause was regional quota, not model code or schema.

No residents were affected; this was synthetic portfolio telemetry.

Change afterwards

Used Azure Functions Flex Consumption for the browser API wrapper.

Kept Azure ML behind a governed layer.

Documented fallback hosting path and evidence checks before claiming readiness.

What I Would Bring To ECC

Azure-native delivery with responsible AI controls by design

Move experiments into governed Azure ML pipelines and model promotion gates.

Deploy through managed online endpoints, batch endpoints, and supportable API boundaries.

Work with Information Governance before real data: DPIA, UK GDPR, access control, retention, and audit.

Coach analysts and data scientists on reproducible, testable MLOps workflows.

Surface monitoring through Power BI-ready outputs that service owners can actually use.

Deployment is not just getting a model behind an endpoint. It is making it reproducible, explainable, monitored, governed, and safe to challenge.

LLM use disclosure

I used an AI assistant to structure and polish the deck.
I verified the content against repository artifacts, model metrics, Azure notes, and screenshots from the running website.